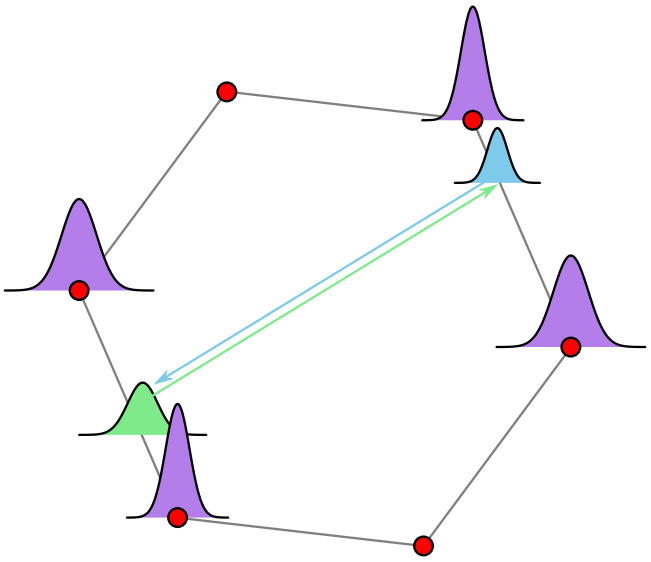
THE UNIVERSITY OF MELBOURNE

Final Report

The Art of Scientific Computation: Electron Wavefunction Approximations and Parallel Computational Methods

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1.1 Electron Wavefunctions

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1.3 Hartree-Fock Approximation

1.4 Many-Body Perturbations

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2 Method

Method text here.

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3.1 Hückel Approximation

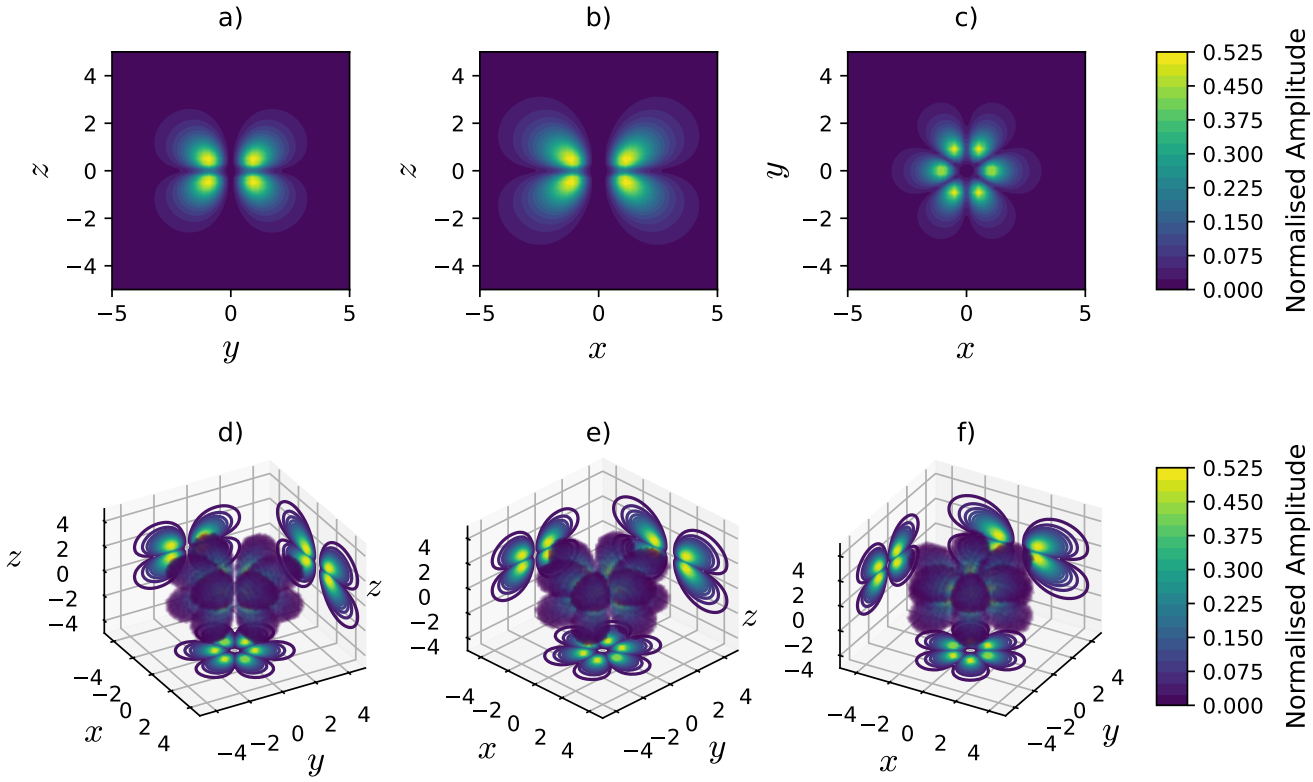


Figure 3.1: Orbital electron wavefunction of Benzene, using the Hückel approximation with orbital angular momentum quantum numbers l=1 and m=0.

3.2 Hartree-Fock Analytical Approximation

3.3 Hartree-Fock Numerical Approximation

3.4 Many-Body Electron Perturbations

3.5 GLSL High-Order Perturbations

4 Discussion

Discussion text here.

5 Conclusion

Conclusion text here.

Appendix A: Additional Information

Appendix B: Program Flowchart

A flowchart is a conceptual aid which highlights input and output that a program should have, as well as the consequences of any decisions that it must make. Sometimes, these will include *pseudocode*, which mimics some of the more common coding structures in a universal way. (For instance, you might say 'loop over all array addresses'.)

This particular flowchart indicates a program that reads three numbers from the terminal, and then produces the largest of the numbers back to the terminal. The colour scheme is not mandatory, but helps the reader to know what is going on at a glance.

This flowchart was drawn with the free 'Lucidcharts' extension to Google Docs, from which the result was exported as an `svg` file, and any unnecessary features were then cropped out with Inkscape. Choose any method you wish to create your own flowchart!

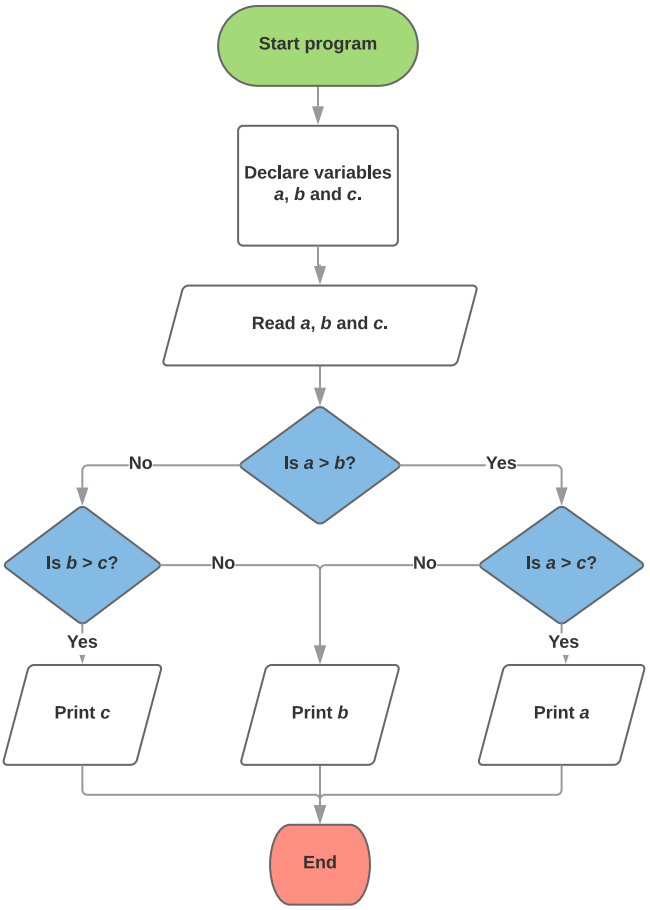


Figure 5.1: A flowchart for the determination of the largest of three user-selected numbers.

Bibliography